

**JKBOSE**

**2026 CLASS 11<sup>TH</sup>**

# **MOTION IN PLANE**

**LECTURE-4**

**PHYSICS**

**WAJAHID SIR (Gabbar Sir)**



# ***Topics To Covered***

- 1** Revision ✓
- 2** Gabbar Numerical Series PYQs
- 3** ..
- 4** ..

Tum Sab Kabil Ho Apne aap  
pe shaq na Karo



# Recap from 1<sup>st</sup> Lecture

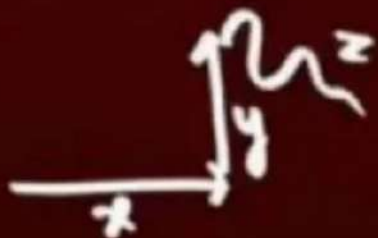
- Motion in 1D



- Motion in 2D



- Motion in 3D



- Displacement in 2D

$$\vec{r} = \vec{x} + \vec{y}$$

- Velocity in 2D

$$\vec{v} = \vec{v}_x + \vec{v}_y$$

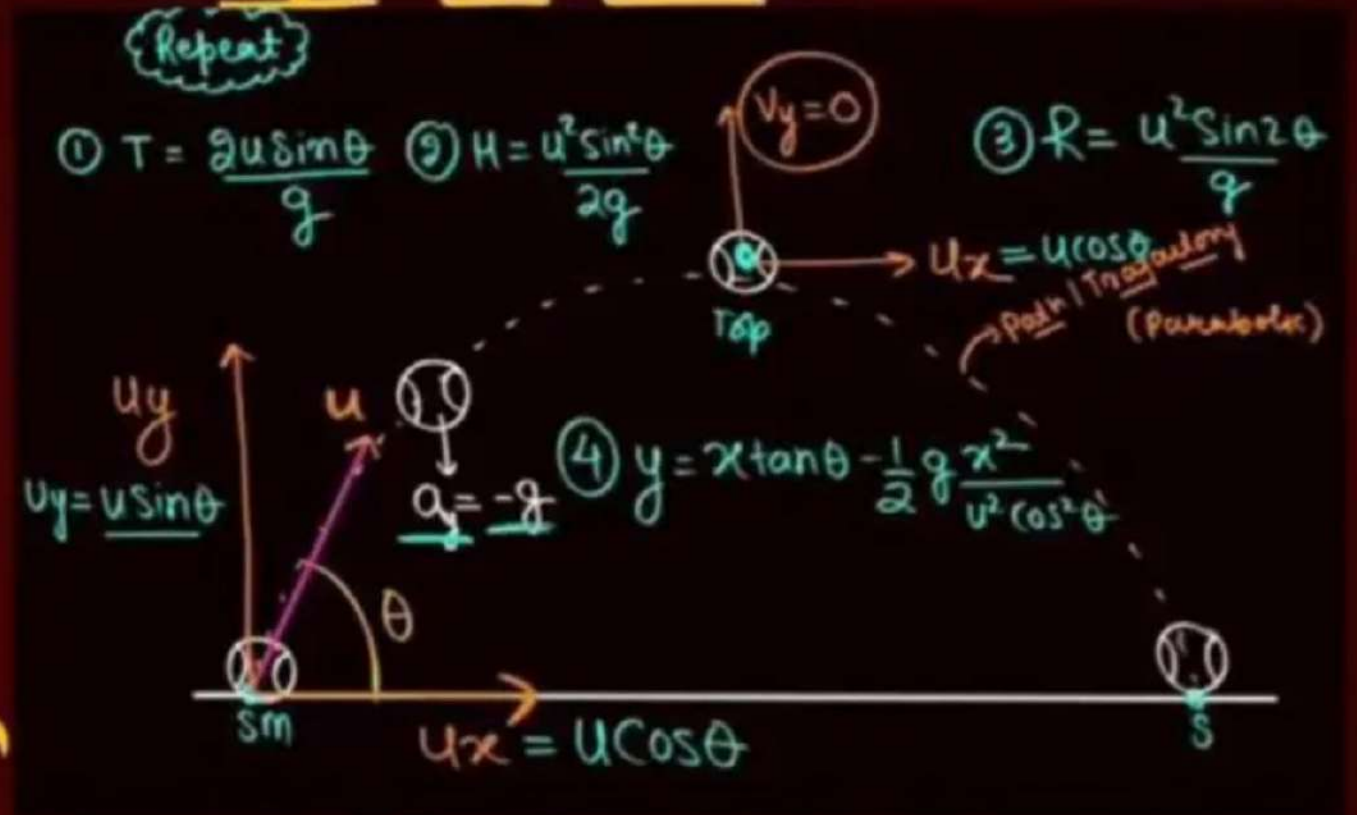
- Acceleration in 2D

$$\vec{a} = \vec{a}_x + \vec{a}_y$$

- Equation of motion

$$\begin{aligned} v &= u + at & S &= ut + \frac{1}{2}at^2 & v^2 &= u^2 + 2as \\ v_x &= u_x + a_x t & S_x &= u_x t + \frac{1}{2}a_x t^2 & v_x^2 &= u_x^2 + 2a_x S_x \\ v_y &= u_y + a_y t & S_y &= u_y t + \frac{1}{2}a_y t^2 & v_y^2 &= u_y^2 + 2a_y S_y \end{aligned}$$

## Projectile Motion





# Projectile Motion

(5-mark) 2026, 2025, 2024, 2023, 2022



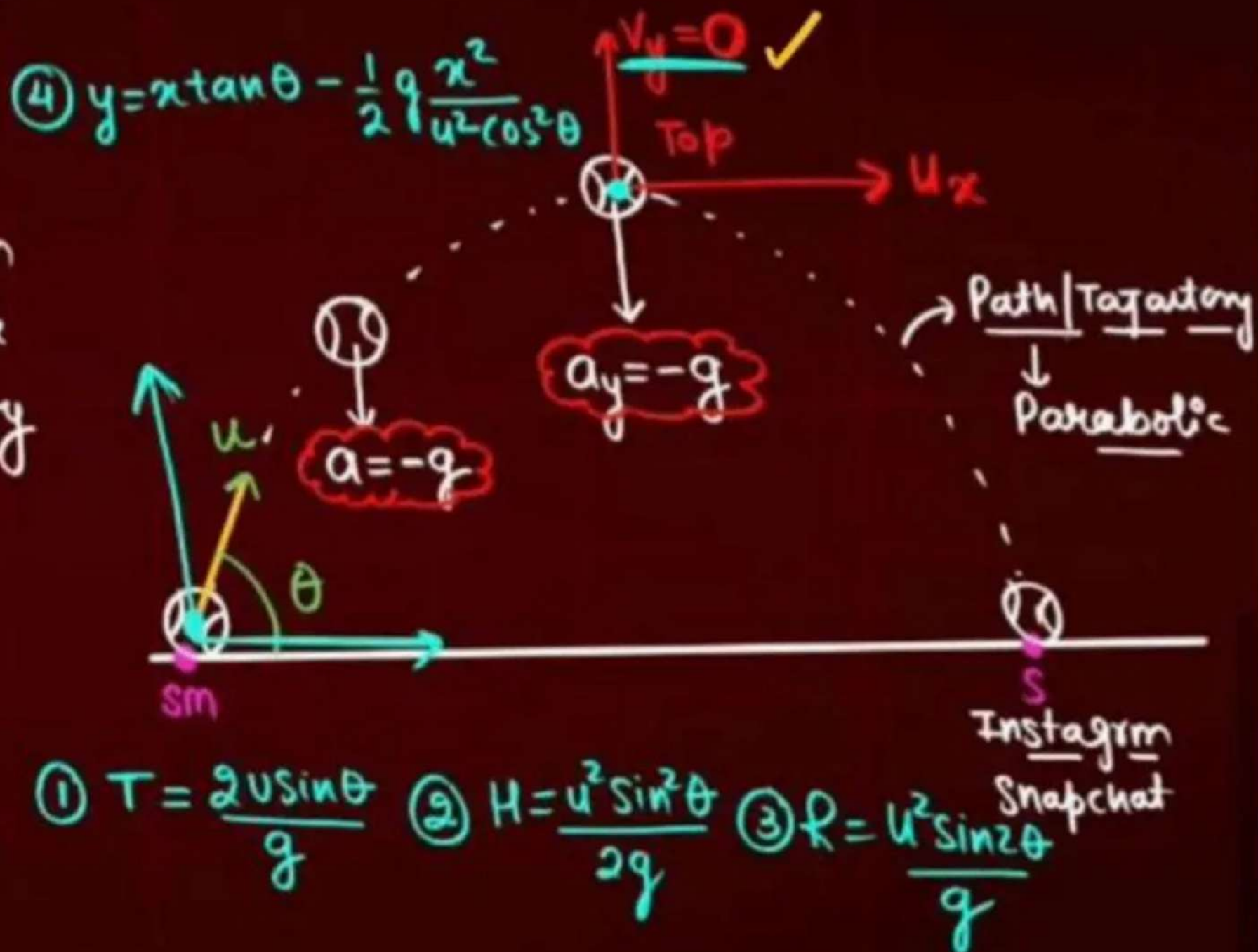
- Example of 2D
- To explain the 2D motion
- when an object is thrown with Some Angle under the influence of gravity

#  $a_y = -g = -9.8 = -10 \text{ m/s}^2$

#  $a_x = 0$

#  $v_y = 0$

# Path | Tangent  $\rightarrow$  Parabolic

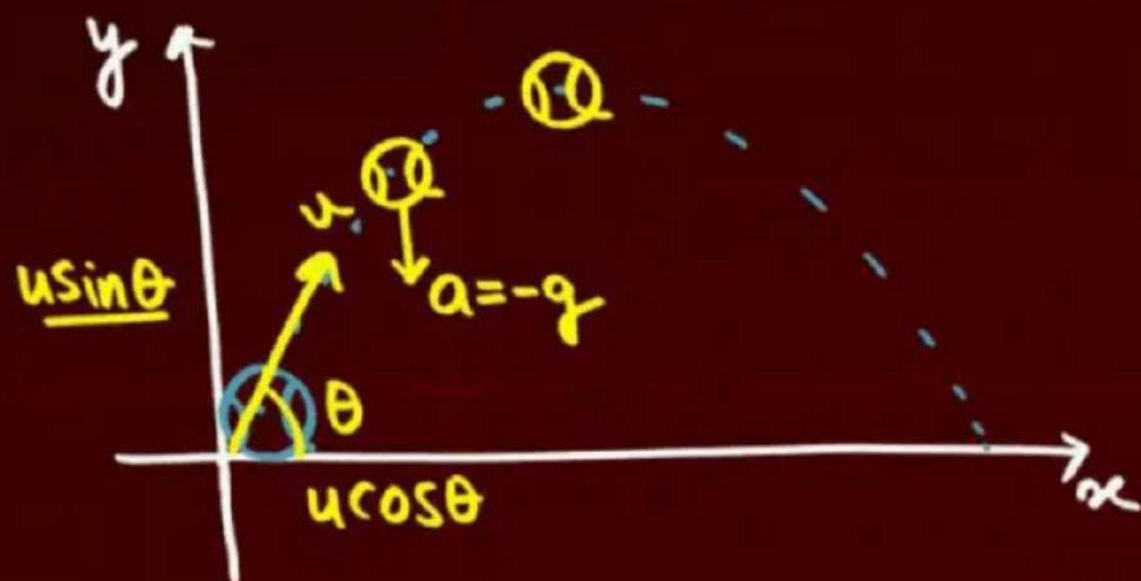




# Recap from 2<sup>nd</sup> Lecture



## Projectile Motion



### ① Time period.

The time for which ball remain in Air  $T = \frac{2u \sin \theta}{g}$

### ② Maximum Height.

The maximum height attain by Projectile

$$H = \frac{u^2 \sin^2 \theta}{2g}$$

### ③ Range

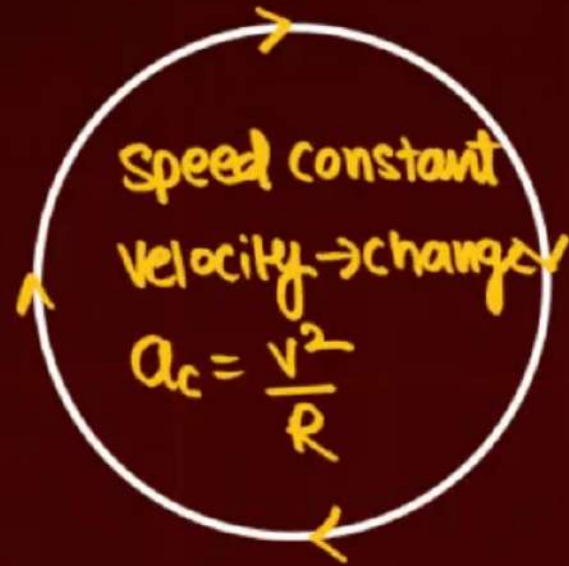
The maximum Horizontal displacement of object  $R = \frac{u^2 \sin 2\theta}{g}$

### ④ Equation of Trajectory $\rightarrow$ Relationship b/w $y$ and $x$

$$y = x \tan \theta - \frac{1}{2} g \frac{x^2}{u^2 \cos^2 \theta}$$

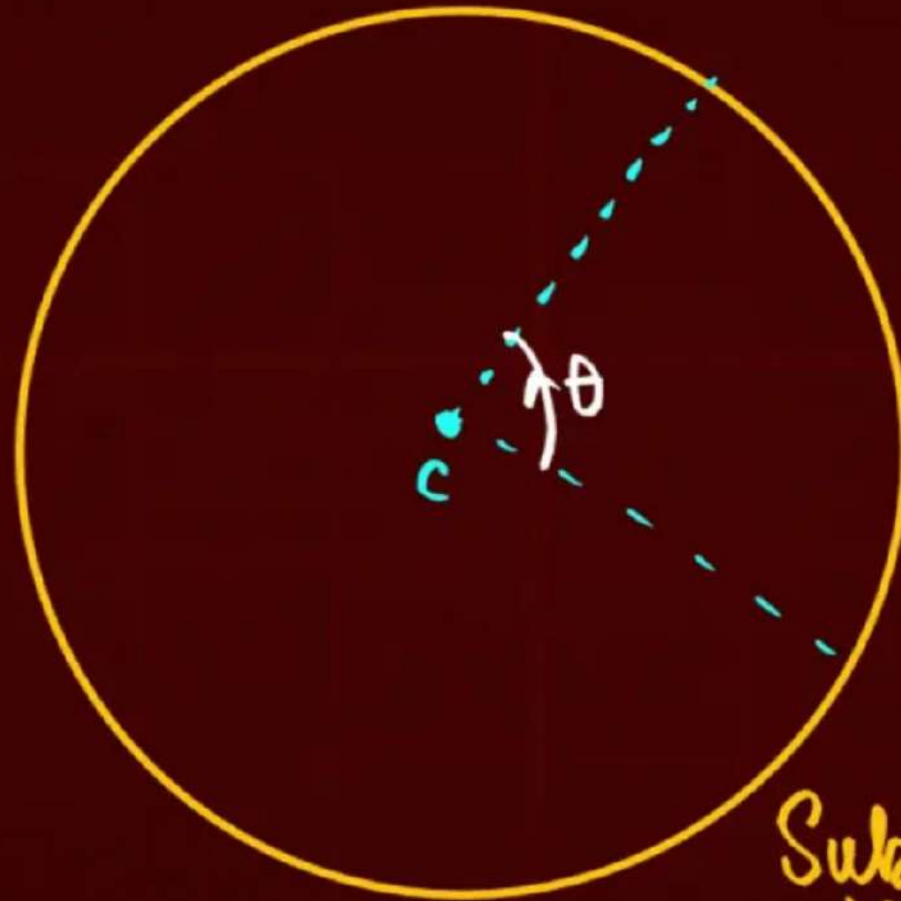
# Recap from 3<sup>rd</sup> Lecture

## Uniform Circular Motion



# Non-uniform Motion

# Speed =  $\frac{2\pi r}{T} = 2\pi r\gamma$



① Angular displacement

$$\theta = \frac{\Delta}{r}$$

$$r\theta = \Delta$$

② angular velocity ( $\omega$ )

$$r\omega = v$$

③ Angular acceleration ( $\alpha$ )

$$r\alpha = a$$

Subtraction of two vector

$$R = \sqrt{A^2 + B^2 - 2AB\cos\theta} \quad \tan\alpha = \frac{B\sin\theta}{A - B\cos\theta}$$

R R



# **Gabbbar**

# **Numerical Series**

# **PYQs**



# QUESTIONS [SUMMER SERIES-A 2026]

A  
B  
C

-5mark

Find (a) time of flight, (b) maximum height, (c) horizontal range of a projectile projected with speed  $v$  making an angle  $\theta$  with horizontal direction from ground.

T

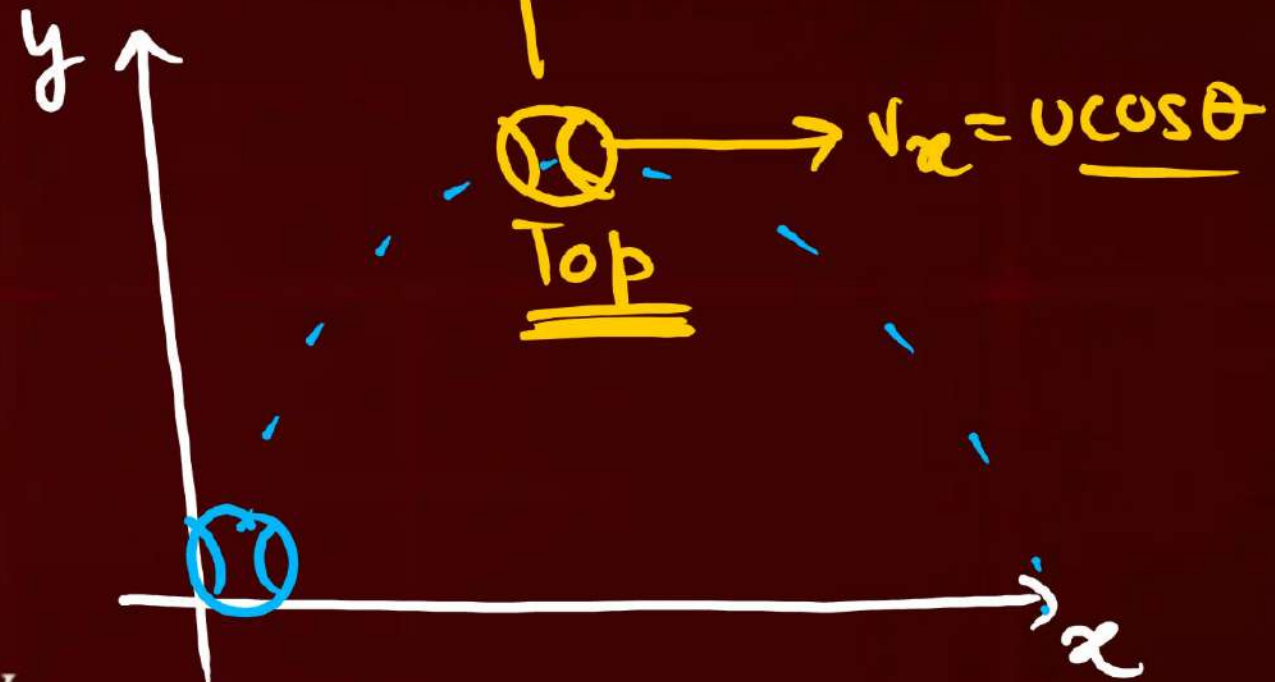
H

R

# QUESTIONS[WINTER SERIES-A 2025]

At what point in its trajectory does a projectile have minimum upward speed.

- ☒ A At the start of the Trajectory
- ☒ B At the end of the Trajectory
- ☒ C At the midpoint of the Trajectory
- ☒ D At the highest point its Trajectory





## QUESTIONS[WINTER SERIES-A 2025]

3mark

✓ Derive an equation of trajectory of projectile fired at an angle.

## QUESTIONS[WINTER SERIES-A 2025]

3mark ✓

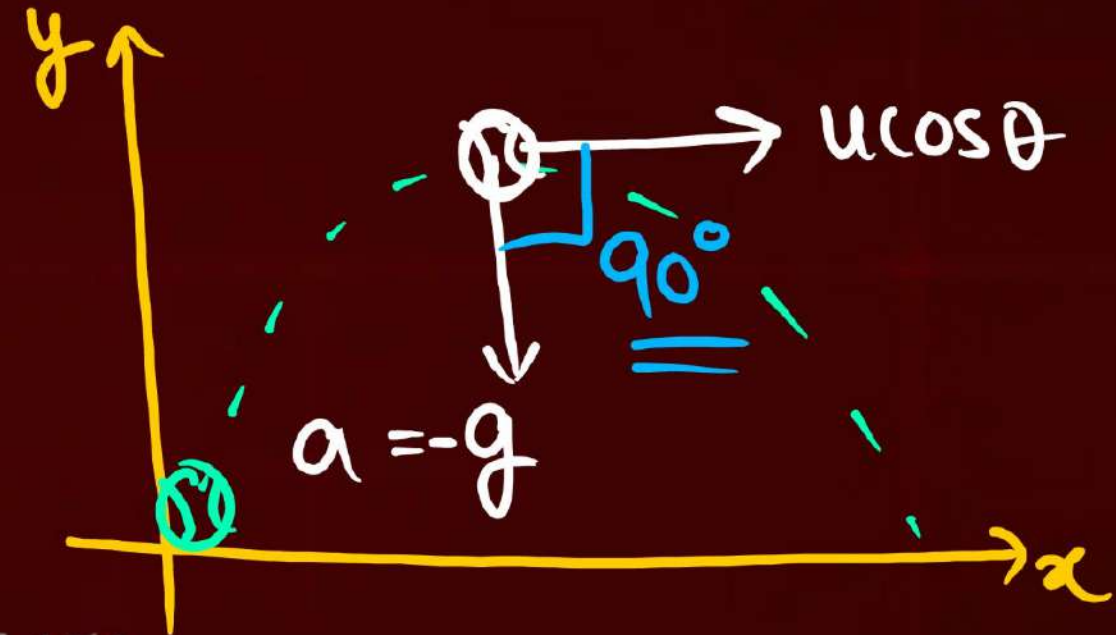
What do you mean by resolution of a vector? Find rectangular components of a vector.



## QUESTIONS[WINTER SERIES-B 2025]

At the top of the trajectory of a projectile, the direction of its velocity and acceleration are:

- A** Parallel to each other
- B** Antiparallel to each other
- C** Inclined to each other at an angle of  $45^\circ$
- D** Perpendicular to each other ✓



## QUESTIONS[WINTER SERIES-C 2025]

The horizontal range covered by the projectile is proportional to:

$$R = \frac{u^2 \sin 2\theta}{g}$$

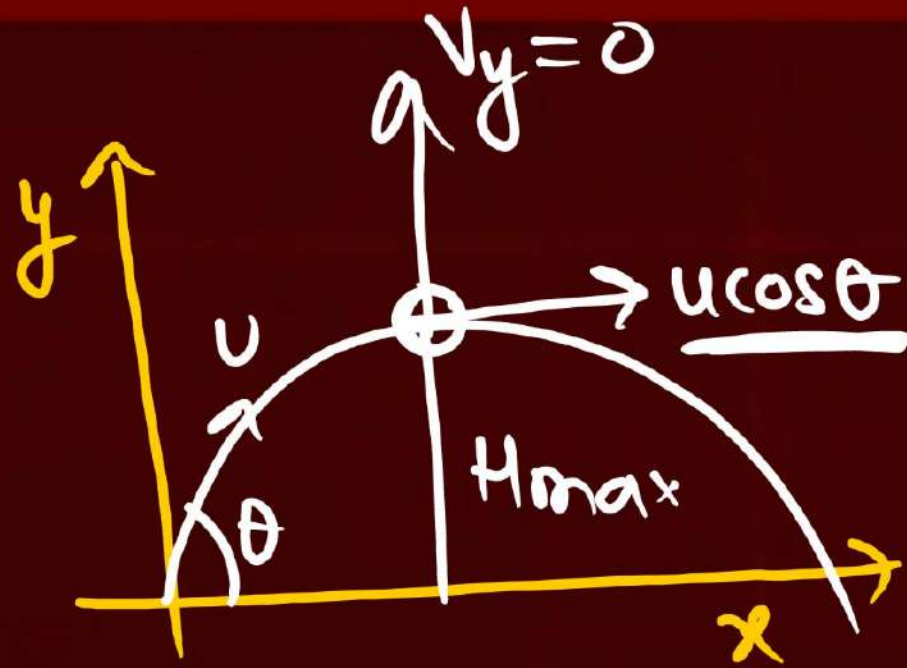
- ☒ A Its velocity
- ☒ B Square of the velocity
- ☒ C Sine of angle of projection
- ☒ D Square of sine of angle of projection



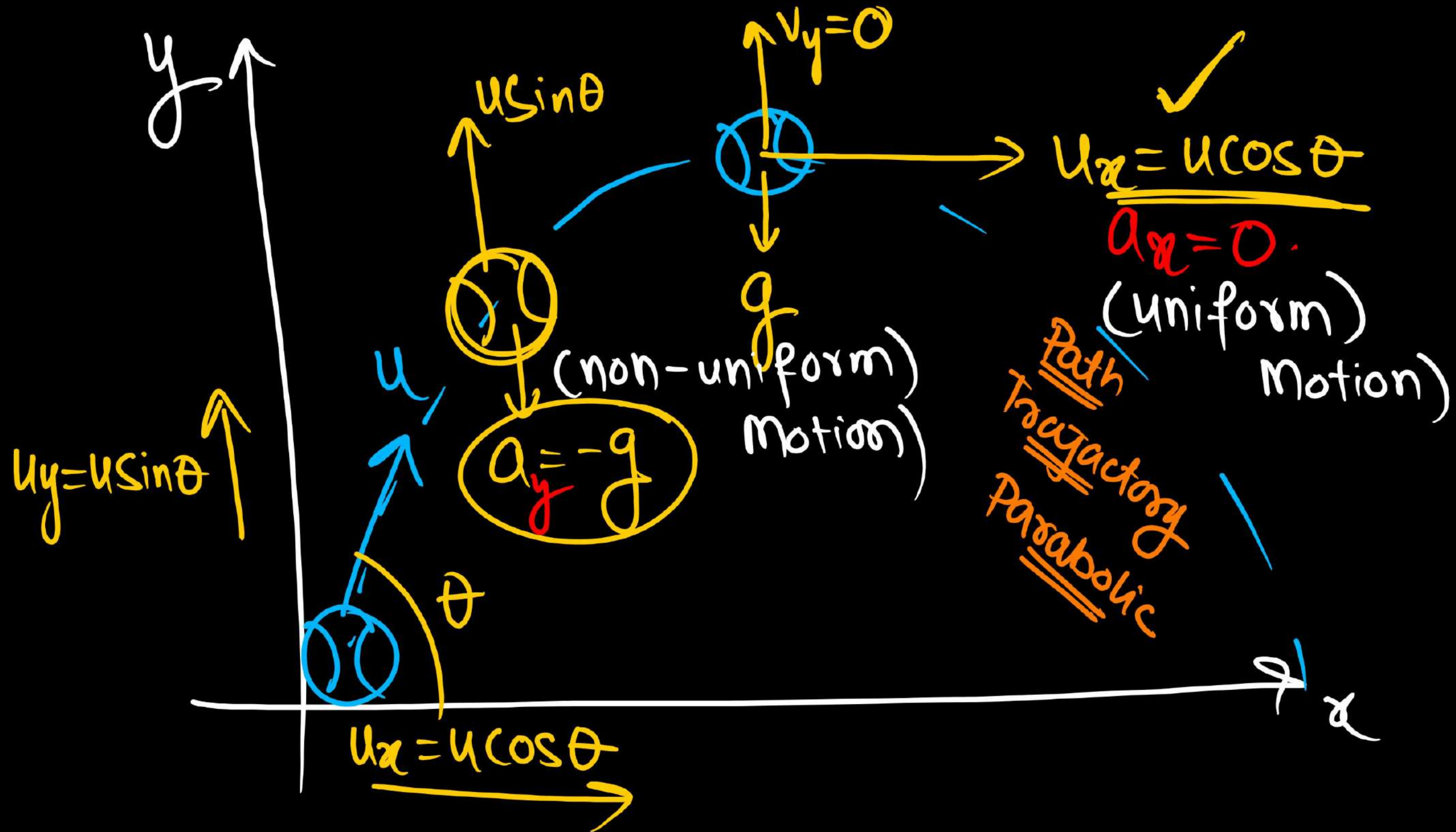
## QUESTIONS[SUMMER SERIES-A 2025]

If a projectile is thrown with velocity  $u$  at an angle  $\theta$  with the horizontal, then velocity at maximum height during projectile motion will be:

- A**  $u \sin \theta$
- B**  $2u \sin \theta$
- C**  $2u \cos \theta$
- D**  $u \cos \theta$







## QUESTIONS [SUMMER SERIES-A 2025]

✓ Show that the path followed by projectile is parabolic when it is projected at an angle  $\theta$  with horizontal.

or

Explain cross product of vectors. (Properties)



# QUESTIONS [SUMMER SERIES-B 2025]

To take the longest possible jump, an athlete should make an angle with horizontal:

$$R = \frac{u^2 \sin 2\theta}{g}$$

$$R_{\max} = \frac{u^2 (\sin 2\theta)^{\max}}{g} \quad 2\theta = 90^\circ$$

$$\theta = \frac{90}{2}$$

$$\theta = 45^\circ$$

$$\sin 2\theta = \max$$

$$\sin 2\theta = 1$$

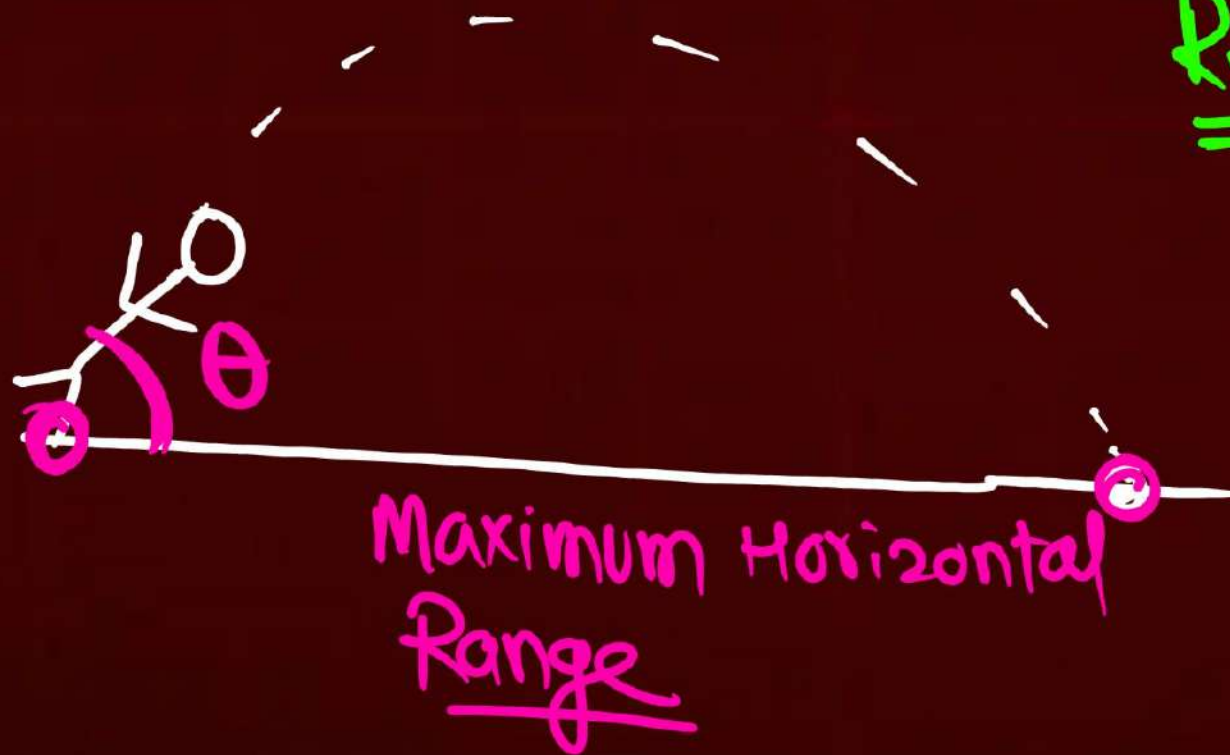
$$\sin 2\theta = \sin 90^\circ$$

A 90°

B 60°

C 45°

D 30°



## QUESTIONS[SUMMER SERIES-C 2025]

The path traced by a projectile in space is known as:



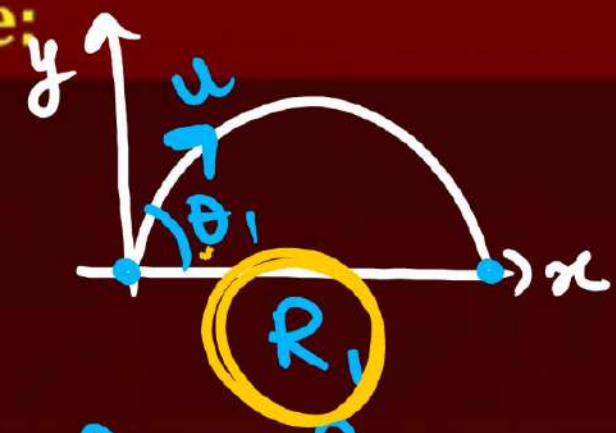
- A** Track
- B** Orbit
- C** ✓ Trajectory
- D** Coral



# QUESTIONS[WINTER SERIES-A 2024]

A projectile will cover same horizontal distance, when the initial angles of projection are:

- ☒ A  $20^\circ, 60^\circ$
- ☐ B  $20^\circ, 70^\circ$
- ☐ C  $30^\circ, 40^\circ$
- ☐ D  $50^\circ, 60^\circ$



$$R_1 = R_2$$

$$\cancel{u^2} \sin 2\theta_1 = \cancel{u^2} \sin 2\theta_2$$

$$\sin 2\theta_1 = \sin 2\theta_2$$

$$\sin 2\theta_1 = \sin 2\theta_2$$



$$\sin 2\theta_1 = \sin 2\theta_2$$

$$\sin x = \sin (180 - x)$$

$$x = 2\theta_1$$

$$180 - x = 2\theta_2$$

$$180 - 2\theta_1 = 2\theta_2$$

11<sup>th</sup> Maths

$$180 = 2\theta_2 + 2\theta_1$$

$$180 = 2(\theta_2 + \theta_1)$$

$$\frac{180}{2} = \theta_2 + \theta_1$$

$$90^\circ = \theta_1 + \theta_2$$

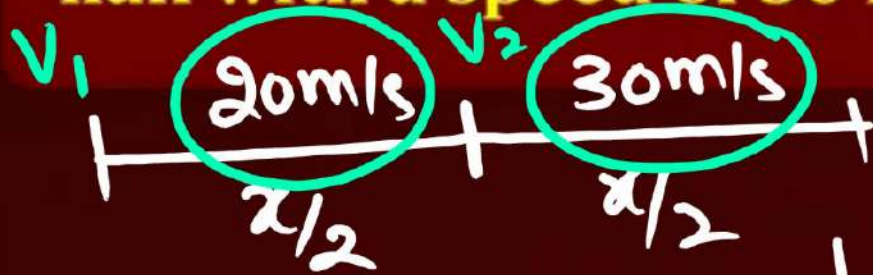


## QUESTIONS[WINTER SERIES-A 2024]

Revision 10



A body covers first half of distance with a speed of 20 m/s and second half with a speed of 30 m/s. Find the average speed of the body.



Sol

$$\frac{2}{V_a} = \frac{1}{v_1} + \frac{1}{v_2}$$

$$\frac{2}{V_a} = \frac{1}{20} + \frac{1}{30}$$

$$V_a = ?$$

$$\frac{2}{V_a} = \frac{30 + 20}{600}$$

$$\frac{2}{V_a} = \frac{50}{600}$$

$$\frac{2}{V_a} = \frac{5}{60}$$

$$\frac{V_a}{2} = \frac{60}{5}$$

$$\frac{V_a}{2} = 12$$

$$V_a = 12 \times 2 = 24 \text{ m/s} \checkmark$$

## QUESTIONS[WINTER SERIES-A 2024]

Define projectile. Derive an expression for maximum height, time of flight and horizontal range of a projectile fired at an angle with the horizontal ✓



## QUESTIONS[SUMMER SERIES-A 2025]

Show that the path followed by projectile is parabolic when it is projected at an angle  $\theta$  with horizontal.

or

Explain cross product of vectors. ✓

## QUESTIONS[JEE MAIN 2025]

The position vector of a moving body at any instant of time is given as  $\vec{r} = (5t^2\hat{i} - 5t\hat{j})\text{m}$ . The magnitude and direction of velocity at  $t = 2\text{s}$  is

- A**  $5\sqrt{17}\text{ m/s}$ , making an angle of  $\tan^{-1} 4$  with -ve Y axis
- B**  $5\sqrt{15}\text{ m/s}$ , making an angle of  $\tan^{-1} 4$  with +ve X axis
- C**  $5\sqrt{17}\text{ m/s}$ , making an angle of  $\tan^{-1} 4$  with +ve X axis
- D**  $5\sqrt{15}\text{ m/s}$ , making an angle of  $\tan^{-1} 4$  with -ve Y axis

$$\alpha = \tan^{-1} 4$$

$$\alpha = \tan^{-1} 4$$



Sol  $\vec{r} = 5t^2\hat{i} - 5t\hat{j}$

$$v = \frac{d\vec{r}}{dt}$$

$$v = \frac{d}{dt}(5t^2\hat{i} - 5t\hat{j})$$

$$v = \frac{d}{dt} 5t^2\hat{i} - \frac{d}{dt} 5t\hat{j}$$

$$v = 5 \frac{dt^2}{dt} \hat{i} - 5 \frac{dt}{dt} \hat{j}$$

$$v = 5 \times 2t^{2-1} \hat{i} - 5 \times 1 t^{1-1} \hat{j}$$

$$\frac{dx^n}{dx} = nx^{n-1}$$

$$v = 10t^1\hat{i} - 5t^0\hat{j}$$

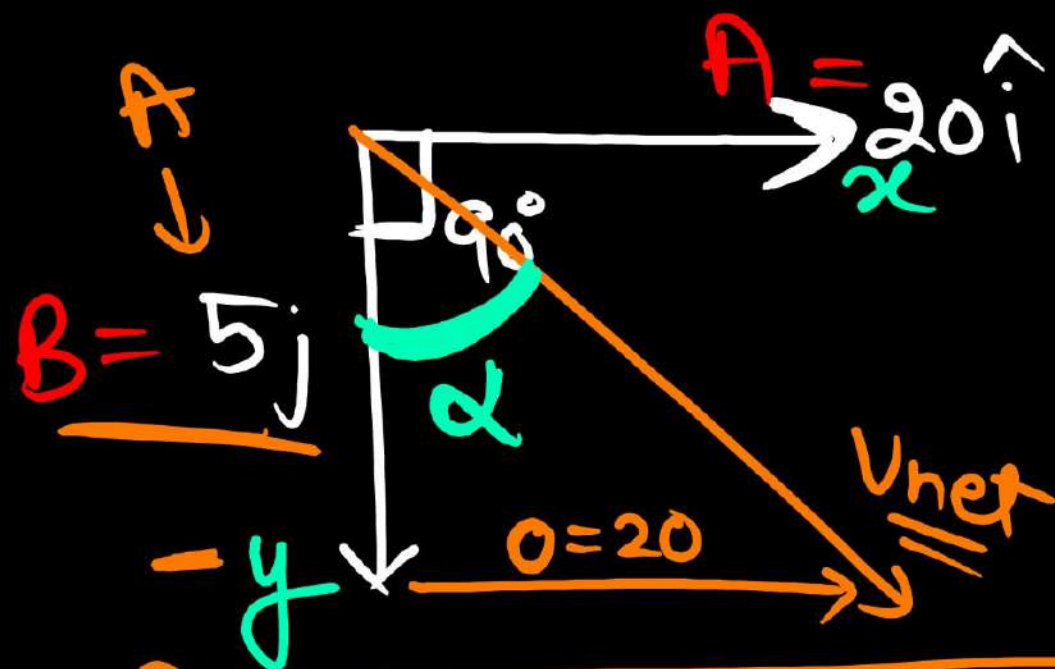
$$v = 10t\hat{i} - 5 \times 1 \times \hat{j}$$

$t = \underline{2\text{sec}}$   $v = 10\underline{t}\hat{i} - 5\hat{j}$

$$v = 10 \times 2\hat{i} - 5\hat{j}$$

$$\boxed{v = 20\hat{i} - 5\hat{j}}$$

$$\underline{V = 20\hat{i} - 5\hat{j}}$$



$$R = \sqrt{A^2 + B^2 + 2AB \cos 90^\circ}$$

$$R = \sqrt{A^2 + B^2 + 2AB \cos 90^\circ}$$

$$R = \sqrt{A^2 + B^2}$$

$$R = \sqrt{(20)^2 + 5^2}$$

$$R = \sqrt{400 + 25}$$

$$R = \sqrt{425}$$

$$R = \sqrt{5 \times 5 \times 17}$$

$$R = \sqrt{5^2 \times 17}$$

$$R = 5\sqrt{17}$$

$$\underline{V_{net} = 5\sqrt{17} \checkmark}$$

y-axis

$$\tan \alpha = \frac{O}{A}$$

$$\tan \alpha = \frac{20}{5} = 4$$

$$\boxed{\tan \alpha = 4}$$

y-axis



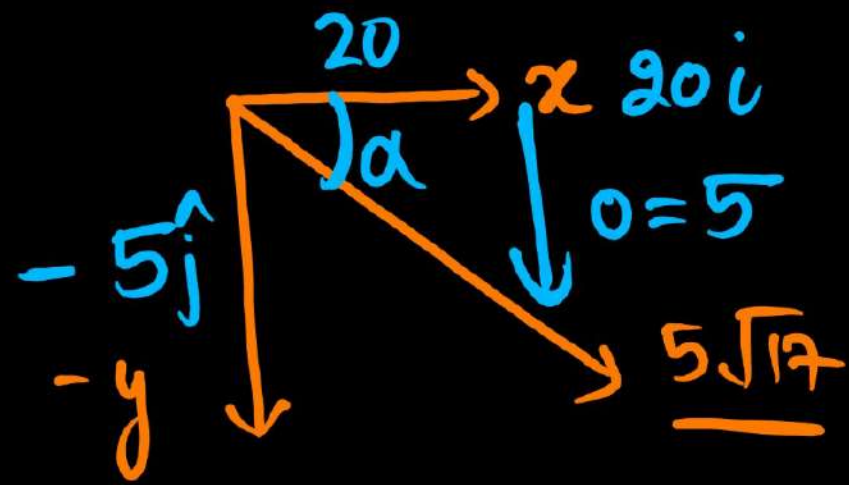


$$\underline{\tan \alpha = 4}$$

$$\alpha = \frac{4}{\tan^{-1}}$$

\*\*

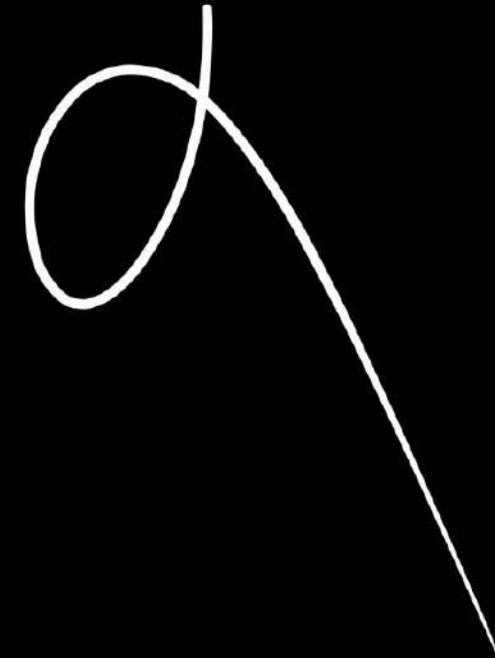
$$\boxed{\alpha = \tan^{-1} 4}$$



$$\tan \alpha = \frac{O}{A}$$

$$\tan \alpha = \frac{5}{20}$$

$$\tan \alpha = \frac{1}{4}$$





## QUESTIONS[NEET 2025]

2021

$$S = \underline{ut} + \frac{1}{2} \underline{at^2}$$
$$\underline{v^2} = v^2 + 2as$$



A particle is given velocity 3 m/s along the positive x-axis subjected to constant acceleration  $4\text{m/s}^2$  along the positive y-axis. Find the displacement of the particle in 2 s.

**A** ✓ 10 m

**B** 8 m

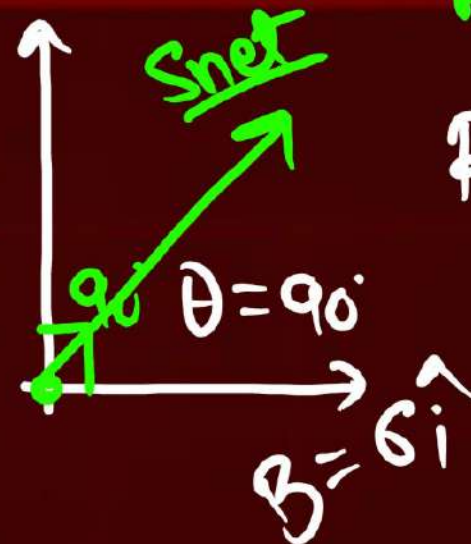
**C** 6 m

**D** 12 m

$$u_x = 3\text{m/s } \hat{i} \rightarrow A = 8\hat{j}$$
$$a_y = 4\text{m/s}^2 \hat{j} \uparrow$$
$$t = 2\text{sec}$$

$$\underline{S} = \underline{ut} + \frac{1}{2} \underline{at^2}$$
$$S = 3\hat{i} \times 2 + \frac{1}{2} 4 \times 2^2$$

$$\underline{S = 6\hat{i} + 8\hat{j}}$$



$$R = \sqrt{A^2 + B^2 + 2AB \cos 90^\circ}$$

$$R = \sqrt{8^2 + 6^2}$$

$$R = \sqrt{64 + 36}$$

$$= \sqrt{100}$$

$$= \sqrt{10^2}$$

$$= 10\text{m}$$



## QUESTIONS[NEET 2023]

Two projectiles A and B are thrown with initial velocities of 40 m/s and 60 m/s at angles  $30^\circ$  and  $60^\circ$  with the horizontal respectively. The ratio of their ranges respectively is ( $g = 10 \text{ m/s}^2$ )

$$\frac{\sin 60}{\sin 120}$$



**A**

4 : 9

**B**

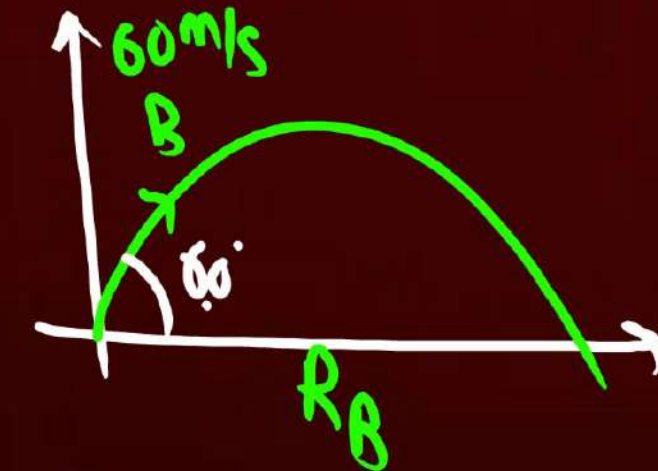
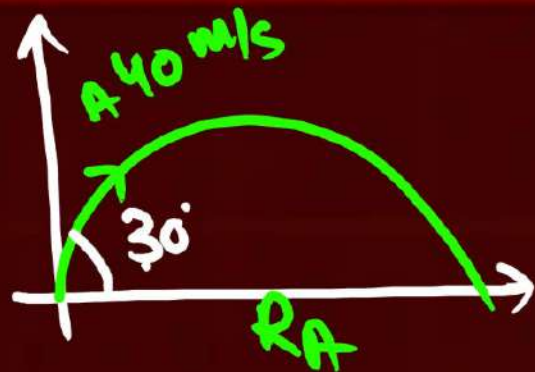
$2 : \sqrt{3}$

**C**

1 : 1

**D**

$\sqrt{3} : 2$



$$\frac{R_A}{R_B} = \frac{U_A^2 \sin 2\theta_A}{U_B^2 \sin 2\theta_B} = \frac{40 \times 40 \sin 2 \times 30}{60 \times 60 \sin 2 \times 60} = \frac{1600 \sin 60}{3600 \sin 120} = \frac{1600 \times \frac{\sqrt{3}}{2}}{3600 \times \frac{\sqrt{3}}{2}} = \frac{1600}{3600} = \frac{4}{9}$$



## QUESTIONS[NEET 2024]



The co-ordinates of a particle moving in x-y plane are given by:  
 $x = 4t$ ,  $y = 3t + 8t^2$ . The motion of the particle is: 2D

- A** Uniform motion along a straight line.
- B** Non-uniformly accelerated.
- C** Uniformly accelerated having motion along a straight line.
- D** Uniformly accelerated having motion along a parabolic path.

$$a_y = 16$$

$$\underline{x = 4t}$$

$$v_x = \frac{dx}{dt} = \frac{d}{dt} 4t$$

$$= 4 \frac{dt^1}{dt}$$

$$= 4 \times 1 \times t^{1-1}$$

$$= 4 \times 1 \times t^0$$

$$= 4 \times 1 \times 1$$

$$= \underline{4 \text{ m/s}}$$

$$a_x = \frac{dv}{dt}$$

$$= \frac{d}{dt} 4$$

$$= \underline{0}$$

$$y = 3t + 8t^2$$

$$v_y = \frac{dy}{dt} = \frac{d}{dt} (3t + 8t^2)$$

$$= \frac{d}{dt} 3t + \frac{d}{dt} 8t^2$$

$$= 3 \frac{dt}{dt} + 8 \frac{d}{dt} t^2$$

$$= 3 + 8 \times 2t^{2-1}$$

$$\underline{v_y = 3 + 16t}$$



$$a_y = \frac{dv_y}{dt}$$

$$= \frac{d}{dt} (\vec{3} + 16t)$$

$$= \frac{d3}{dt} + \frac{d}{dt} 16t$$

$$= 0 + 16 \frac{d}{dt} 1$$

$$\boxed{a_y = 16}$$

const acc

uniform accelerated motion

$$\rightarrow x = 4t$$

$$\checkmark \boxed{V_x = 4} \checkmark$$

$$\checkmark \boxed{a_x = 0}$$

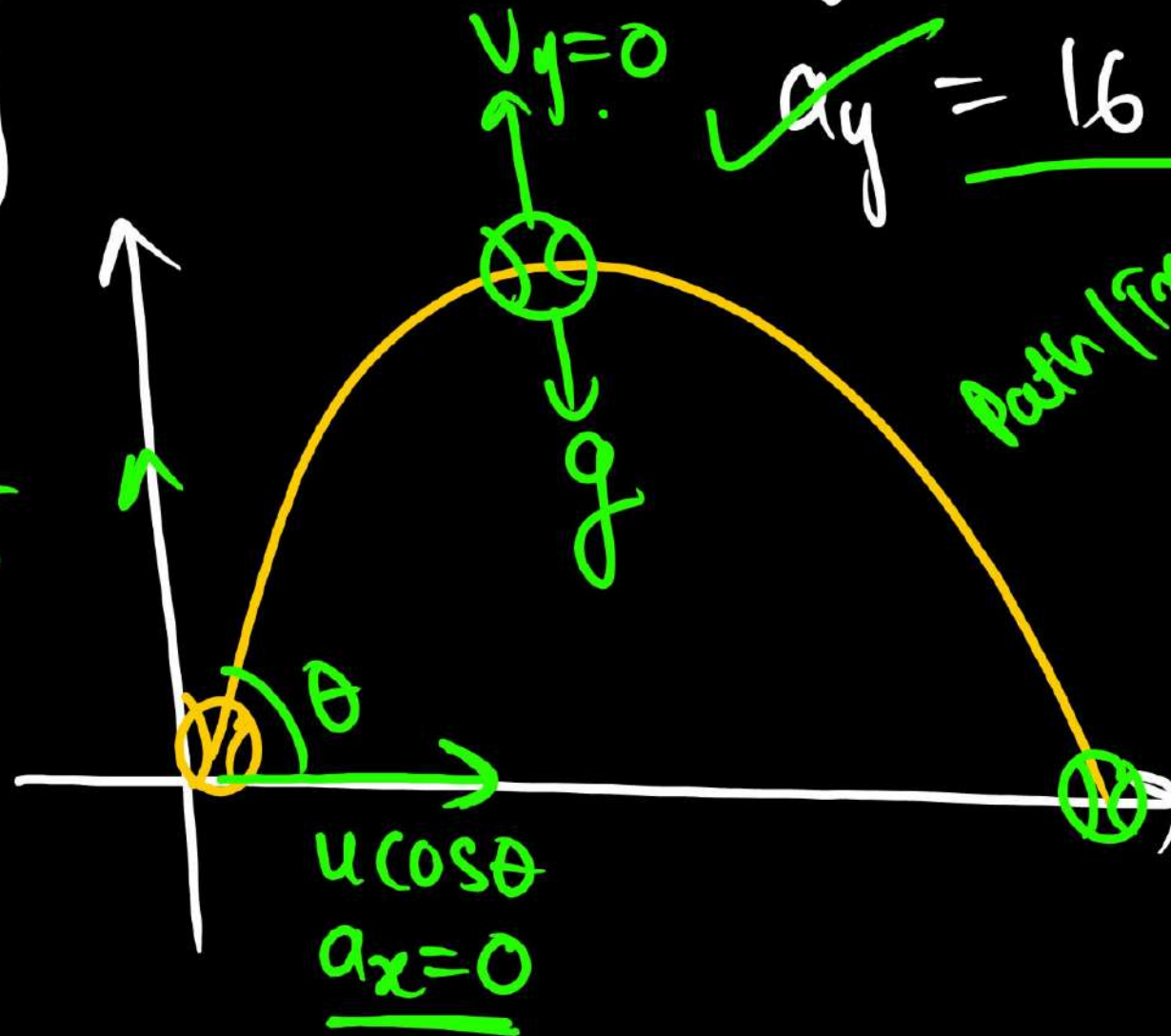
$$\uparrow y = 3t + 8t^2$$

$$\checkmark V_y = 3 + 16t$$

$$\checkmark a_y = 16$$

path/trajectory  
parabolic

$$\underline{u \sin \theta}$$





## Jadugar Question

- ✓ State parallelogram law of vector addition and derive an equation for resultant of two vectors.  $(R) \& \tan \alpha$
- Define cross product of two vectors and write its properties.
- Define dot product of two vectors and write its properties.
- What is projectile motion, Define trajectory. Derive equations for
  - (a) Time of flight  $T$
  - (b) Maximum height  $H$
  - (c) Range when the projectile is thrown with some initial velocity  $R$
  - (d) Equation of trajectory

## QUESTIONS[NEET 2019]

4/10



The position vector of particle changes with time according to the relation  $\vec{r} = 15t^2\hat{i} + (4 - 20t^2)\hat{j}$  Find the magnitude of the acceleration at  $t = 1$

- A** 50
- B** 25
- C** 40
- D** 100



## SUMMARY

Numericals → PYQ (Boards)  
→ Neet  
→ Jee.

chapter complete



**Thank**  
*You*

Revision → 10 min

Notes → There's a  
late